PassFrame tuition

Ratio & Proportion

Year 4 Pt 2 — Homework ⏱ 45 min

Worked example — read this through before you start

Worked example: Share £50 in the ratio 2:3.

Method & answer: Total parts = 5. £50 ÷ 5 = £10 per part. 2 parts = £20, 3 parts = £30.

Easy / Standard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. Write the ratio 8:12 in simplest form.

2. Write the ratio 15:20 in its simplest form. A) 3:4 B) 3:5 C) 5:4 D) 1:5

3. There are 4 cats and 12 dogs. Write the ratio of cats to dogs in simplest form.

4. Share £30 in the ratio 1:5. Select ALL statements that are TRUE: A) The shares are £5 and £25 B) The shares are £15 and £15 C) There are 6 parts in total D) The larger share is £25 E) There are 5 parts in total

5. Share 40 sweets in the ratio 3:5.

6. Write the ratio 14:21 in simplest form.

7. For every 3 red counters there are 4 blue counters. If there are 9 red counters, how many blue counters are there? A) 9 B) 10 C) 12 D) 16

8. A recipe for 2 people needs 80g of rice. How much is needed for 6 people?

9. Write the ratio 10:4 in simplest form.

10. Share £60 in the ratio 1:2.

11. A map scale is 1 cm : 4 km. How many km does 5 cm represent?

12. Write the ratio 18:24 in its simplest form. A) 2:3 B) 3:4 C) 6:8 D) 9:12

13. There are 6 boys and 9 girls. Write the ratio of boys to girls in simplest form.

14. Share 36 marbles in the ratio 1:3. Select ALL statements that are TRUE: A) The shares are 9 and 27 B) The shares are 18 and 18 C) There are 4 parts in total D) The larger share is 27 E) There are 3 parts in total

15. A recipe for 4 people needs 200g of flour. How much is needed for 8 people?

Standard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. Share £84 in the ratio 2:3:2.

2. A recipe for 4 people needs 240g of rice. How much is needed for 10 people? A) 500g B) 550g C) 600g

6. Share 63 sweets in the ratio 2:5.

7. A model train is built to a scale of 1:30. If the model is 6 cm long, how long is the real train? A) 36cm B)

11. The ratio of cats to dogs at a shelter is 3:5. There are 32 animals in total. Select ALL statements that are TRUE: A) There are 12 cats B) There are 8 parts in total C) There are 20 dogs D) There are 16 cats E) 1 part is 3 animals

12. Simplify the ratio 56:24. A) 7:3 B) 3:7 C) 8:3 D) 28:12

D) 650g

90cm C) 180cm D) 210cm

3. The ratio of red to blue paint is 2:7. If there are 10 litres of red, how much blue is there?

8. For a squash drink, the ratio of concentrate to water is 1:6. If 200 ml of concentrate is used, how much water is needed?

13. A paint mix uses red and white in the ratio 1:5. How much white paint is needed for 4 litres of red?

4. Simplify the ratio 32:40.

9. Share £120 in the ratio 3:5.

14. Share 90 counters in the ratio 4:5.

5. A class has boys to girls ratio 5:6. There are 25 boys. Select ALL statements that are TRUE: A) There are 30 girls B) 1 part is 5 pupils C) There are 11 parts in total D) There are 24 girls E) 1 part is 6 pupils

10. A recipe for 5 people needs 150 ml of cream. How much is needed for 8 people?

15. A scale drawing uses 1 cm to represent 5 m. How many cm represent 35 m?

Standard / Advanced — *work through together, answer 5 in the lesson* *15 min (1 min per question)*

1. Simplify the ratio 36:48, then share £70 using the simplified ratio. A) 3:4, shares = £30 and £40 B) 4:3, shares = £40 and £30 C) 3:4, shares = £35 and £35 D) 2:3, shares = £28 and £42

6. A fruit punch is made from orange, pineapple and mango juice in the ratio 3:2:1. If the jug holds 1,800 ml in total, how much of each juice is used?

11. In a garden, the ratio of roses to tulips to daisies is 4:3:5. If there are 45 daisies, how many flowers are there in total? A) 96 B) 100 C) 104 D) 108

2. Write the ratio 15:9 in its simplest form, then state the larger share if a total of 24 is shared in this ratio. A) 5:3, larger share = 15 B) 5:3, larger share = 9 C) 3:5, larger share = 15 D) 5:4, larger share = 12

7. A recipe for 6 people needs 360g of flour and 3 eggs. How much flour and how many eggs (to the nearest half egg) are needed for 15 people?

12. A recipe uses butter and flour in the ratio 1:3. If 450g of flour is used, how much butter is needed, and what is the combined weight of butter and flour?

3. A ratio of 2:5 is equivalent to what ratio with a total of 21?

8. The ratio of teachers to pupils on a trip is 2:15. If there are 8 teachers, select ALL statements that are TRUE: A) There are 60 pupils B) There are 68 people on the trip in total C) There are 30 pupils D) There are 76 people on the trip in total E) 1 part is 4 people

13. Two numbers are in the ratio 5:8. Their difference is 45. Select ALL statements that are TRUE: A) The numbers are 75 and 120 B) The numbers are 70 and 115 C) 1 part is 15 D) 1 part is 9 E) There are 3 parts in the difference

4. Two numbers are in the ratio 3:4 and their sum is 63. What is the larger number? A) 27 B) 28 C) 36 D) 42

9. A sum of money is shared between Priya and Sam in the ratio 4:7. Sam receives £60 more than Priya. How much does Priya receive, and what is the total sum shared?

14. A car travels 252 miles using 28 litres of fuel. At the same rate, how much fuel is needed to travel 360 miles?

5. A map scale is 1:100,000. Find the real distance, in km, for 4.5 cm on the map.

10. A scale model of a bridge is built at 1:200. The model is 45 cm long. How long is the real bridge, in metres?

15. A rectangle has length to width ratio 7:3. If the perimeter is 100 cm, find the length and width.

PassFrame Tuition · Ratio & Proportion · Year 4 Pt 2 Homework · 45 questions, 15 from each difficulty band, 45 min total · Mark scheme kept by tutor